

Literals in Java

A literal is a fixed value that is written directly in a Java program.

In simple words, whatever value we write directly in the code is called a literal.

Example

```
System.out.println(100);
```

Here, **100** is a literal.

Types of Literals in Java

1. Integer Literal

Whole numbers without decimal points are called Integer Literals.

```
System.out.println(100);
```

```
System.out.println(5000);
```

Output

```
100  
5000
```

2. Floating Point Literal

Numbers containing decimal points are called Floating Point Literals.

```
System.out.println(99.99);  
System.out.println(45.67);
```

Output

```
99.99  
45.67
```

3. Character Literal

A single character enclosed within single quotes (' ') is called a Character Literal.

```
System.out.println('A');  
System.out.println('Z');
```

Output

```
A  
Z
```

4. String Literal

A group of characters enclosed within double quotes (" ") is called a String Literal.

```
System.out.println("Java");
```

```
System.out.println("Hello Students");
```

Output

```
Java  
Hello Students
```

5. Boolean Literal

The values `true` and `false` are called Boolean Literals.

```
System.out.println(true);  
System.out.println(false);
```

Output

```
true  
false
```

Printing Different Types of Literals

```
public class LiteralDemo {  
  
    public static void main(String[] args) {  
  
        System.out.println(100);  
  
        System.out.println(99.99);  
  
        System.out.println('A');  
  
        System.out.println("Java");  
  
        System.out.println(true);  
    }  
}
```

Output

```
100  
99.99  
A  
Java  
true
```

- 100 → Integer Literal
- 99.99 → Floating Point Literal
- 'A' → Character Literal
- "Java" → String Literal
- true → Boolean Literal

Special Characters in Java

Sometimes we want to print text in a new line, give space, or print double quotes inside a message. For this, Java provides some special characters called **Escape Sequences**.

1. \n (New Line)

\n is used to move the cursor to the next line.

Example

```
public class Demo {  
    public static void main(String[] args) {  
        System.out.println("Java\nPython");  
    }  
}
```

Output

Java

Python

Explanation

- Java is printed on the first line.
- \n moves the cursor to the next line.
- Python is printed on the second line.

2. \t (Tab Space)

\t is used to give a tab space between two values.

Example

```
public class Demo {  
    public static void main(String[] args) {  
        System.out.println("Name\tAge");  
    }  
}
```

Output

Name Age

Explanation

- \t creates extra horizontal space.
- It helps in formatting output neatly.

3. \" (Double Quote)

\" is used to print double quotes inside a string.

Example

```
public class Demo {  
    public static void main(String[] args) {  
        System.out.println("\"Java\"");  
    }  
}
```

Output

"Java"

Explanation

- Double quotes normally start and end a string.
- To print them as part of the output, use \".

Using Multiple Escape Sequences Together

Example

```
public class Demo {  
    public static void main(String[] args) {  
        System.out.println("Name\t:\tRahul\nCourse\t:\tJava");  
    }  
}
```

Output

```
Name      :      Rahul  
Course    :      Java
```

Explanation

- \t provides spacing.
- \n moves to the next line.

String Concatenation

Joining two or more strings together is called **String Concatenation**.

In Java, the + operator is used for concatenation.

Example 1

```
public class Demo {  
    public static void main(String[] args) {  
        System.out.println("Hello " + "Students");  
    }  
}
```

Output

```
Hello Students
```

Explanation

- "Hello " and "Students" are joined together using +.

Example 2

```
public class Demo {  
    public static void main(String[] args) {
```

```
        System.out.println("Java " + "Programming");  
    }  
}
```

Output

Java Programming

Concatenation with Different Literals

```
public class Demo {  
    public static void main(String[] args) {  
        System.out.println("Age : " + 25);  
        System.out.println("Grade : " + 'A');  
        System.out.println("Passed : " + true);  
    }  
}
```

Output

Age : 25
Grade : A
Passed : true